Independent Personal Research Report (Non-degree, Non-thesis)

One-hour-ahead PM10/PM2.5 prediction, spatial vulnerability assessment, and 2026 near-future scenario analysis using public air-quality data in Daejeon, Korea

An integrated case study of time-series forecasting, spatial vulnerability, and near-future scenarios based on public monitoring data

It documents an independent research project personally conducted by the author, outside any official supervision or peer-review process.

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TABLE OF CONTENTS

Table of Contents

Abstract

1. Introduction
  1.1 Background and need for the study
  1.2 Literature review
  1.3 Research gap
  1.4 Research objectives and research questions

2. Study area and data
  2.1 Study area
  2.2 Air−quality data
  2.3 Meteorological data
  2.4 Local PM alerts and dust advisories
  2.5 Derived data tables

3. Methodology
  3.1 Overall workflow
  3.2 Data processing
  3.3 Feature engineering

3.4 Machine learning models
3.5 Spatial analysis with QGIS
3.6 Summary statistics & Power BI dashboards
3.7 Multi−analog scenario 2026 & ensemble risk

4. Results
4.1 RQ1 – One−hour−ahead prediction performance
4.2 RQ2 – Spatial vulnerability in 2021 – 2023
4.3 2026 multi−analog scenarios & ensemble risk

5. Discussion
5.1 Predictive capability and limitations
5.2 Spatial vulnerability and policy implications
5.3 Strengths, limitations and future work

Conclusion

References

ABSTRACT

One-hour-ahead PM10/PM2.5 prediction, spatial vulnerability assessment, and 2026 near-future scenario analysis using public air-quality data in Daejeon, Korea

Particulate matter (PM10) and fine particulate matter (PM2.5) are key air pollutants that are closely linked not only to short-term aggravation of respiratory and cardiovascular diseases but also to long-term chronic illness and premature mortality. As awareness of these health impacts has increased, both the international community and Korea have strengthened air-quality standards and implemented various policy measures such as high-concentration alerts and seasonal management schemes. In this context, city-scale air-quality management has gained importance, and there is growing demand for practical analytical frameworks that combine short-term prediction, spatial vulnerability assessment, and screening of near-future risk levels.

This study focuses on Daejeon Metropolitan City and presents a local case study that integrates, within a single end-to-end pipeline relying only on public monitoring data, (i) one-hour-ahead prediction of PM10 and PM2.5 concentrations, (ii) spatial vulnerability analysis at the administrative-dong and East/West sector levels for 2021–2023, and (iii) construction of multi-analog scenarios in which the observed patterns in 2021, 2022, and 2023 are shifted to 2026, followed by ensemble vulnerability assessment. To this end, hourly time-series data are constructed by combining PM observations from the AirKorea urban air-monitoring network in Daejeon with ASOS meteorological data, local PM alert records, and national dust and fine-dust advisories. Random Forest and XGBoost models for one-hour-ahead prediction are then trained using lagged and moving-average features of PM and meteorological variables.

Using 2021–2022 data for model training and 2023 (validation) and 2024 (external validation) for evaluation, the models generally achieve coefficients of determination (R^2) on the order of 0.8–

0.9 for both PM10 and PM2.5. In contrast, prediction errors increase during high-concentration periods associated with local PM alerts and national dust advisories, indicating that high-pollution episodes currently represent a key vulnerability of the feature set and modeling approach.

For spatial analysis, QGIS and Python are used to aggregate hourly PM concentrations and exposure indicators at the administrative-dong level for 2021–2023, and to assess vulnerability by combining WHO and Korean annual air-quality standards with local PM alerts and dust-advisory information. The results show that several administrative dongs, including Gwanpyeong-dong, Munpyeong-dong, Noeun-dong, Dunsan-dong, Munchang-dong, and Jeongnim-dong, act as structurally high-risk areas with both elevated mean PM10/PM2.5 levels and high proportions of high-concentration hours. At the broader scale, the eastern sector of Daejeon (Dong-gu, Jung-gu, and Daedeok-gu) exhibits generally higher mean concentrations and higher high-episode shares than the western sector (Seo-gu and Yuseong-gu).

For the near-future analysis, we construct multi-analog scenarios (Scenarios A/B/C) by treating each of the years 2021, 2022, and 2023 as a plausible pattern for 2026, and use the trained RF/XGBoost models to predict PM10 and PM2.5 distributions in 2026. We then compute, across the three scenarios, two models, and two pollutants, ensemble high-episode frequencies and counts of high-risk class occurrences. Under the PM2.5 criterion, all urban air-monitoring stations are classified as high-risk from the ensemble perspective, whereas no station is classified as high-risk under the PM10 criterion. This suggests that the Daejeon urban monitoring network is structurally exposed to a high level of risk in terms of PM2.5.

This study demonstrates a reproducible analysis framework that integrates short-term prediction, spatial vulnerability assessment, and near-future scenario analysis using only public monitoring data, while jointly considering PM10 and PM2.5. The resulting framework can serve as a basis for identifying priority management zones in urban air-quality policy, refining local alert systems, and supporting more advanced modeling studies. Future work should extend the temporal coverage, enrich the feature set, apply the framework to additional cities, and explore hybrid

approaches that couple data-driven models with physically based air-quality and climate–emission scenarios.

Keywords: **PM10; PM2.5; short-term forecasting; spatial vulnerability; scenario analysis; Daejeon; machine learning; QGIS; Power BI; public data**

1. Introduction

1.1. Background and need for the study

Particulate matter (PM10) and fine particulate matter (PM2.5) have small aerodynamic diameters and can penetrate deeply into the respiratory tract. Short-term exposure is associated with exacerbation of respiratory and cardiovascular diseases, increases in emergency room visits, and elevated mortality, whereas long-term exposure is known to cause severe health impacts such as chronic lung disease, cardiovascular disease, and premature death [14,15]. As recognition of these health effects has grown, the international community and domestic authorities have strengthened air-quality management standards and implemented various policy efforts, including high-concentration alert systems [14,16–18].

In 2021, the World Health Organization (WHO) revised its Air Quality Guidelines, tightening both annual and short-term standards for PM10 and PM2.5 [16]. In Korea, the enactment of the Special Act on the Reduction and Management of Fine Dust, the implementation of seasonal management policies, and the introduction of various emission reduction measures have expanded the institutional foundation for reducing high-concentration fine dust episodes [17,18]. However, the ultimate implementation unit of air-quality management is the individual city, and because emission characteristics, topography, meteorology, and population distribution differ across cities, fine-scale analysis and tailored management strategies at the city level are required.

City-scale air-quality management can be approached along three major axes. First, short-term prediction can be used to forecast concentration changes over the next few hours to a day and to support alert issuance and behavioral guidance. Second, spatial vulnerability assessment can be used to identify areas that are structurally exposed to higher concentrations and higher levels of population exposure. Third, near-future scenarios and policy scenarios can be used to examine how risk levels may change in the coming years. The aim of this study is to connect these three axes within a single pipeline using only public monitoring data.

1.2 Literature review

The related literature can be broadly categorized into four main strands of research.

First, numerous short-term prediction studies have been conducted for single stations or a small number of stations. Using deep learning models such as LSTM, GRU, and CNN, as well as tree-based machine-learning models such as Random Forest and XGBoost, researchers have predicted one-hour-ahead or 1-day-ahead PM10 and PM2.5 concentrations at specific monitoring sites [1–4]. These studies typically use past PM concentrations and meteorological variables (e.g., temperature, relative humidity, wind speed, wind direction, and precipitation) as input features, and compare models based on prediction performance metrics such as RMSE, MAE, and R^2. However, relatively few studies extend their prediction results into administrative-unit spatial vulnerability analyses or interpret them in terms of exposure patterns linked with alerts and dust advisories.

Second, there is a body of work using physically based three-dimensional numerical models (e.g., CMAQ, WRF-Chem) at regional or national scales [5–7]. These models combine meteorological fields, emission inventories, and chemical reaction modules, and are widely used to evaluate long-term average concentrations and the effects of policy scenarios (such as emission reductions). While they have the advantage of explicitly representing physical and chemical processes and allowing a wide range of scenario simulations, the configuration and computational cost of such models are substantial. As a result, the barrier to entry is high for practitioners who wish to construct reproducible analysis frameworks for a single city using only public monitoring data.

Third, many GIS-based studies have focused on producing static maps of spatial concentration distributions [8–10]. Typical examples include visualizing annual or seasonal mean PM levels by administrative unit and mapping exceedances of regulatory standards. However, these studies often emphasize long-term averages, and relatively few attempts have been made to integrate temporal variability, local PM alerts and dust advisories, and error patterns of machine-learning predictions into a unified spatial vulnerability analysis. In particular, studies that simultaneously consider both PM10 and PM2.5 when defining vulnerability remain limited [8–10].

Fourth, there are studies that evaluate future concentrations under climate and emission scenarios [11–13]. These works combine climate scenarios such as RCPs/SSPs with emission scenarios to evaluate future changes in mean concentrations and exceedance frequencies (e.g., for 2030 or 2050). While such studies are useful from a long-term and large-scale perspective, the number of examples that integrate short-term prediction, spatial vulnerability, and near-future scenario analyses into a single pipeline, linking public monitoring data with machine-learning models at the city scale, is relatively small.

1.3 Research gap

Considering the above literature, the research gap addressed in this study can be summarized as follows:

(1) Machine-learning-based short-term prediction studies have largely focused on comparing prediction performance at single stations or a small number of stations. Few examples systematically link predicted values and performance metrics with spatial vulnerability analyses at the administrative-dong or sector level [1–4].

(2) Physically based three-dimensional numerical models are powerful tools for regional and national policy evaluation, but they require substantial data and computational resources. This makes it burdensome to build reproducible analysis systems targeting a single city using only public monitoring data. There remains a need for lightweight, data-driven pipelines that are accessible at the practitioner level.

(3) Studies based on static mean maps provide intuitive visualization of average concentration levels and standard exceedances at specific times or over specific periods, but they do not sufficiently incorporate temporal variability, exposure during local PM alerts and dust advisories, or differences in the performance of machine-learning prediction models. Spatial vulnerability analyses that jointly consider these aspects, especially for both PM10 and PM2.5, remain limited [8–10].

(4) Future concentration assessments based on climate and emission scenarios are valuable at long temporal scales and broad spatial scales, but the approach of treating "recent years' observed patterns as possible near-future realizations" through multi-analog and ensemble techniques is relatively new [11–13]. There are few studies that assume the past three years (2021–2023) as possible patterns for a near-future year (2026) and combine them with machine-learning prediction models for vulnerability assessment.

Accordingly, the contributions of this study can be summarized in three points:

(1) We propose an integrated framework that uses public PM and meteorological observations and simple time-series features to predict one-hour-ahead PM10 and PM2.5 and directly links predicted values and error patterns with spatial vulnerability analyses at the administrative-dong and sector levels.

(2) We define vulnerability indicators that jointly consider PM10 and PM2.5 at the administrative-dong and East/West sector levels, and carry out multi-dimensional evaluations that combine mean concentrations, standard exceedance ratios, and exposure during local PM alerts and dust advisories.

(3) We construct multi-analog scenarios by shifting the observed patterns in 2021–2023 to 2026, and perform ensemble analyses that integrate RF/XGBoost models and both pollutants. This allows us to identify local areas that consistently exhibit high risk across diverse near-future patterns, providing a local case study of spatially persistent vulnerability.

1.4 Research objectives and research questions

The overarching objective of this study is to implement, for Daejeon Metropolitan City and using only public monitoring data, an end-to-end pipeline that integrates:

(i) one-hour-ahead prediction of PM10 and PM2.5,

(ii) spatial vulnerability analysis at the administrative-dong and East/West sector levels for 2021–2023, and

(iii) multi-analog scenarios and ensemble vulnerability assessment in which the 2021–2023 observed patterns are shifted to 2026.

To this end, we set the following three research questions:

RQ1. Using publicly available PM and meteorological data and simple time-series features, can we reliably predict one-hour-ahead PM10 and PM2.5 concentrations at 11 urban air-monitoring stations in Daejeon, even when the year and conditions change?

To address this question, we train models on 2021–2022 data and evaluate R^2, RMSE, and MAE for PM10 and PM2.5 separately in 2023 (validation) and 2024 (external test). We also compare model performance under different conditions, including normal periods, local PM alert hours, national dust advisory periods, and high-concentration events, to quantitatively analyze model vulnerabilities and generalization capability.

RQ2. How are mean concentrations, standard exceedance ratios, and exposure during local PM alerts and dust advisories spatially distributed across administrative-dong and East/West sectors in Daejeon during 2021–2023?

To answer this, we compute, for each administrative dong, three-year mean PM10 and PM2.5, exceedance status relative to WHO and Korean annual standards, mean PM and cumulative exposure time during local PM alerts and dust advisories, and East/West sector-level summary statistics. Based on these, we identify structural hotspots that exhibit simultaneously high vulnerability for both PM10 and PM2.5.

RQ3. If the observed patterns in 2021–2023 were to reoccur in 2026, do there exist administrative dongs or sectors that remain consistently vulnerable, showing high concentrations of both PM10 and PM2.5 across all three shifted scenarios (A/B/C) regardless of the specific year used?

For this purpose, we construct Scenario A, B, and C by mapping 2021, 2022, and 2023, respectively, onto 2026, and use the trained RF/XGBoost models to predict PM10 and PM2.5 distributions. For each administrative dong, we compute scenario-specific mean concentrations, time fractions exceeding thresholds (e.g., PM10 $\geq$ 50 μg/m³, PM2.5 $\geq$ 25 μg/m³), and ensemble high-episode ratios that jointly consider the three scenarios and two models. Based on these, we detect administrative units and sectors that repeatedly emerge as high-risk across diverse future patterns.

The remainder of this report is organized as follows.

Chapter 2 describes the study area and datasets used. Chapter 3 explains the methodology, including the data-processing pipeline, feature engineering, Random Forest/XGBoost models, QGIS-based spatial analysis, Power BI dashboards, and construction of the 2026 multi-analog scenarios. Chapter 4 presents results corresponding to the three research questions (RQ1–RQ3). Chapter 5 provides an integrated discussion of these results, including limitations and policy implications. Finally, Chapter 6 summarizes the main conclusions and outlines directions for future research and practical application.

2. Study area and data

2.1 Study area

Daejeon Metropolitan City is a mid-sized city located in the inland central region of the Republic of Korea, where urban, residential, commercial, and industrial areas coexist. This study focuses on 11 urban air-monitoring stations within Daejeon. The locations of these stations are based on coordinates provided by AirKorea and were checked in a GIS environment together with administrative-dong boundaries (Figure 1).

Figure 1

Administrative units are grouped into eastern and western sectors. The eastern sector comprises Dong-gu, Jung-gu, and Daedeok-gu, while the western sector comprises Seo-gu and Yuseong-gu. This classification reflects differences in urban structure, emission patterns, major transport corridors, and the distribution of industrial and residential areas. In general, the eastern sector has higher shares of industrial, logistics, and old downtown areas, whereas the western sector is characterized by residential, educational, research, and office functions. To reflect these spatial differences, subsequent analyses consider both administrative-dong and East/West sector-level indicators.

2.2 Air-quality data

PM data are obtained from monthly finalized datasets for the Daejeon urban air-monitoring stations provided by AirKorea. The raw data consist of PM10 and PM2.5 concentrations by station and hour. Monthly Excel files are collected and reorganized into a single time-series format to construct the raw_pm_final data table (Table 1).

The main characteristics of raw_pm_final are as follows:

Time period	2021-01-01 01:00 to 2024-12-31 23:00
Temporal resolution	1 hour
Spatial units	11 urban air-monitoring stations in Daejeon
Key variables	ts_kst (Korea Standard Time), station_id, pm10, pm25

Table 1. Summary of raw_pm_final

(PM data from urban air-monitoring stations in Daejeon)

During data construction, duplicate records at the same time stamp were removed. Obvious outliers and sensor errors were either removed or treated as missing values based on quality checks. Short gaps were filled using limited interpolation, whereas long gaps were excluded from the analysis.

2.3 Meteorological data

Meteorological data are obtained from the ASOS (Automated Surface Observing System) stations operated by the Korea Meteorological Administration. Observations of temperature, relative humidity, wind speed, wind direction, and precipitation at ASOS stations near Daejeon are collected and used to construct the raw_weather_final data table (Table 2).

The main characteristics of raw_weather_final are:

Time period and resolution	2021-01-01 01:00 to 2024-12-31 23:00, 1-hour intervals
Key variables	ts_kst, temperature, relative_humidity, wind_speed, wind_dir, precipitation, etc.

Table 2. Summary of raw_weather_final (meteorological data from ASOS stations)

All meteorological data are aligned to the same time axis as the PM data using ts_kst. The PM prediction models are then designed to use both meteorological variables and PM time series as input features to explain concentration variability.

2.4 Local PM alerts and dust advisories

Local PM alerts and national dust/fine-dust advisories are compiled based on publicly available information from the Ministry of Environment and the Korea Meteorological Administration.

Using the history of alerts and advisories issued when PM10 or PM2.5 exceeded certain thresholds, we construct the local_alert_events_pm10 and local_alert_events_pm25 data tables. Each table includes variables such as ts_kst, area (East/West sector), and y_loc_pm10 or y_loc_pm25 (alert level or binary label). Reliable local alert data are available for the period 2021–2023.

Dust and fine-dust advisories at the national level are organized into the national_alert_events_daejeon data table, which includes dust_stage (dust-stage label). This table indicates whether a given time stamp falls within a dust advisory or warning period, and the dust_stage label is later used in processed_with_dust for analysis.

2.5 Derived data tables

Based on the raw data described above, several derived data tables are constructed for analysis (Table 3).

pm_with_alerts	This table is created by merging raw_pm_final with local_alert_events_pm10, local_alert_events_pm25, and national_alert_events_daejeon. For each time and station, it adds the labels y_loc_pm10, y_loc_pm25, and dust_stage.
processed_final	This table is obtained by joining pm_with_alerts with raw_weather_final on ts_kst, followed by missing-data and outlier handling. Basic time features such as year, month, dayofweek, hour, and is_weekend are added. processed_final serves as the basis for feature engineering and model training.
processed_with_dust	This table extends processed_final by including dust and alert labels (dust_stage, y_loc_pm10, y_loc_pm25) explicitly for analysis of alert and dust situations and scenario construction.
predictions_full_rf_xgb	This table stores the predicted one-hour-ahead PM10 and PM2.5 concentrations and residuals for 2021–2024, obtained by applying Random Forest and XGBoost models.
processed_with_preds_both	This final master table is constructed by merging processed_with_dust with predictions_full_rf_xgb using station_id and ts_kst. It includes observed values, targets, predictions, errors, and alert/dust labels, and is used as the primary dataset for Power BI dashboards and the 2026 scenario analysis.

Table 3. Summary of derived data tables and their roles.

The time coverage of all derived data tables is harmonized to 2021-01-01 01:00 through 2024-12-31 23:00.

3. Methodology

3.1 Overall workflow

The overall data pipeline for the analysis proceeds as follows:

raw_pm_final, raw_weather_final

→ pm_with_alerts (merging alert and dust labels)

→ processed_final (basic preprocessing and creation of time features)

→ processed_with_dust (integrating alert and dust labels)

→ feature engineering (creation of lag, rolling, and meteorological lag features)

→ RF/XGBoost model training and prediction

→ predictions_full_rf_xgb (full-period predictions for 2021–2024)

→ processed_with_preds_both (integration of observations, predictions, and labels)

Subsequently, processed_with_preds_both is combined with spatial data (stations, daejeon_umd, daejeon_ew) to produce QGIS analysis layers. Python scripts are used to compute summary tables, which are then imported into Power BI to construct dashboards. Finally, multi-analog scenarios that shift 2021–2023 patterns to 2026 are generated, and the trained RF/XGBoost models are reused to perform 2026 predictions and ensemble vulnerability analyses.

3.2 Data processing

All time-series data are ordered by ts_kst (Korea Standard Time). raw_pm_final and raw_weather_final are joined using ts_kst and station_id (or ASOS station ID) as keys. When temporal mismatches arise between PM and meteorological records, we permit merging only when the nearest time stamp falls within ±1 hour.

Local alert and dust-advisory data are merged into processed_final using ts_kst to generate the labels y_loc_pm10, y_loc_pm25, and dust_stage. Time stamps without label entries are treated as "no alert" or "no dust event".

For missing-value treatment, short gaps in PM concentrations are filled using limited linear interpolation based on adjacent time steps, while long gaps or periods with simultaneous missing values in multiple variables are excluded from analysis. For meteorological variables, missing values are filled using either interpolation between neighboring time steps or Last Observation Carried Forward, as appropriate.

3.3 Feature engineering

Based on processed_with_dust, target variables are defined as the one-hour-ahead PM10 and PM2.5 concentrations, denoted target_pm10 and target_pm25. These are obtained by shifting the pm10 and pm25 time series forward by 1 hour along ts_kst.

Input features are constructed as follows (Table 4):

Time features	year, month, dayofweek, hour, is_weekend, etc.
PM time-series features	pm10_lag1–pm10_lag3, pm10_roll_mean_3h, pm10_roll_mean_6h, pm10_roll_mean_12h, pm10_roll_mean_24h; pm25_lag1–pm25_lag3, pm25_roll_mean_3h, pm25_roll_mean_6h, pm25_roll_mean_12h, etc.
Meteorological lag features	temperature_lag1, relative_humidity_lag1, wind_speed_lag1
Alert and dust indicators (optional)	y_loc_pm10, y_loc_pm25, dust_stage (converted to dummy variables when used)

Table 4. List of main features used for PM10/PM2.5 prediction.

During the creation of lag and rolling features, initial time steps with incomplete histories are excluded from analysis. After target generation, all features are aligned to the same index to maintain temporal consistency between predictors and targets.

3.4 Machine learning models

For short-term prediction, we employ tree-based ensemble models: Random Forest and XGBoost (a gradient boosting model). Both can flexibly capture non-linear relationships and interactions among variables and are relatively amenable to interpretation.

The temporal split for model training and evaluation is as follows (Table 5):

Training period	2021–2022 (full years)
Validation period	2023 (full year)
External test period	2024 (full year)

Table 5. Data split for model training, validation, and external testing

Separate models are trained for PM10 and PM2.5. For the Random Forest model, hyperparameters such as n_estimators, max_depth, and min_samples_leaf are tuned; for XGBoost, key hyperparameters such as learning_rate, max_depth, subsample, and colsample_bytree are explored. Hyperparameter optimization is performed using RMSE and R^2 on the validation year (2023), and final settings are provided in the Appendix.

After training, the models are fitted on the training period (2021–2022), and predictions are generated for the validation period (2023) and the external test period (2024). For assessing generalization performance across multiple years, predictions are also generated for the full 2021–2024 period and stored in predictions_full_rf_xgb.

3.5 Spatial analysis with QGIS

Spatial analyses are conducted using QGIS and GeoPandas. First, station coordinates (dmX, dmY in WGS84) from stations.csv are used to create point layers representing the monitoring stations. These coordinates are then transformed from EPSG:4326 to EPSG:5186.

Next, the Daejeon administrative-dong boundary layer (daejeon_umd) is loaded, transformed to the same EPSG:5186 coordinate system, and used to spatially join each station to its corresponding administrative dong (Figure 2)

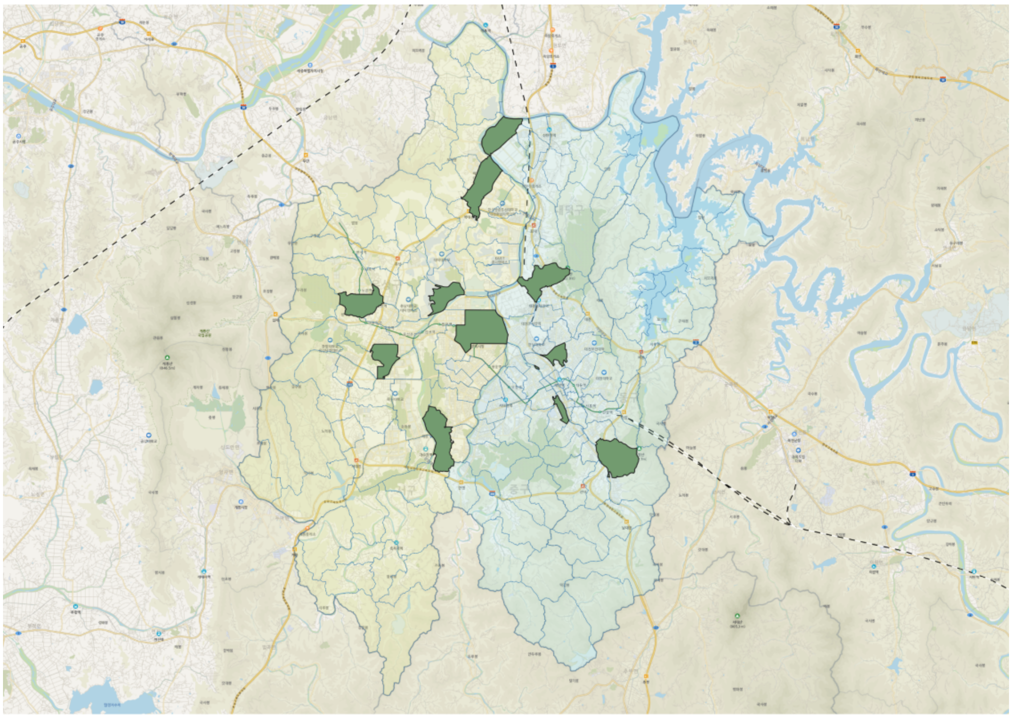


Figure 2. Spatial join of monitoring stations with administrative-dong boundaries in Daejeon.

From processed_with_preds_both, the 2021–2023 period is extracted, and station- and time-specific PM10/PM2.5 observations and predictions are reformatted into an intermediate table, for_qgis_2021_2023. This table is combined with station locations to generate stations_time_2021_2023.gpkg, which is used to construct station-level time-series animations.

To compute administrative-dong-level averages, for_qgis_2021_2023 is spatially joined with the administrative-dong boundaries, and PM10/PM2.5 means are aggregated at the ts_kst × administrative-dong level to create umd_time_pm_2021_2023.gpkg. For mapping three-year mean PM concentrations and standard exceedances, additional summary layers umd_pm_stats_2021_2023_pm10.gpkg and umd_pm_stats_2021_2023_pm25.gpkg are generated, which contain administrative-dong-level means and exceedance ratios. East/West sector information is inserted using daejeon_ew.gpkg.

3.6 Summary statistics & Power BI dashboards

Using Python-based summary scripts, several tables are generated that contain statistics by administrative dong, sector, and condition (Table 6). The main tables are:

bi_admin_pm_stats_2021_2023	Administrative-dong-level mean PM10/PM2.5, WHO and Korean standard exceedance ratios, and high-concentration time fractions.
bi_area_ew_stats_2021_2023	East/West sector-level mean concentrations and exceedance ratios.
bi_model_perf_2021_2024	Prediction performance (RMSE, MAE, R^2) by year, season, time of day, and condition, for each model (RF/XGB) and pollutant (PM10/PM2.5).
bi_alert_pm10_alert, bi_alert_pm25_alert, bi_alert_dust	Summary of exposure time, mean concentration, and maximum concentration during local PM alert periods and dust advisory periods.

Table 6. Main summary tables for Power BI dashboards**.**

These tables are imported into Power BI to construct a dashboard with the following page structure (Table 7):

Overview	Summary of annual and seasonal PM levels and model performance.
Spatial	Visualization of vulnerability indicators at the administrative-dong and East/West sector levels.
Alerts & Dust	Exposure patterns during local PM alerts and dust advisory periods
Model performance	Detailed comparison of model performance by pollutant and condition.
Scenario	Scenario-based predictions for 2026 using environmental data from 2021, 2022, and 2023 as analogs.

Table 7. Structure of Power BI dashboard pages.

The dashboard presents key numerical and spatial patterns in tables and graphs, and serves as a major reference for the results described in Chapter 4.

3.7 Multi-analog scenario 2026 & ensemble risk

To analyze near-future vulnerability in 2026, we treat each of the years 2021, 2022, and 2023 as an analog year. First, using processed_final, we construct scenario-specific feature sets for each pollutant: scenario_2026_features_pm10_A/B/C and scenario_2026_features_pm25_A/B/C. In these files, PM, meteorological, and time features are mapped so that the resulting time series share the same month, day-of-week, and hour structure as 2026.

Next, the trained RF and XGBoost models are reused to generate predictions of one-hour-ahead PM10 and PM2.5 for each scenario. The prediction results are stored in scenario_2026_preds_pm10_rf_xgb and scenario_2026_preds_pm25_rf_xgb.

For scenario aggregation at the administrative-dong and East/West sector levels, we combine each scenario's predictions with station locations and daejeon_umd to calculate mean concentrations and time fractions exceeding thresholds (e.g., PM10 $\geq$ 50 μg/m^3, PM2.5 $\geq$ 25 μg/m^3) for each dong and sector. From this, we generate bi_admin_pm_stats_2026_pm10, bi_admin_pm_stats_2026_pm25, bi_area_ew_stats_2026_pm10, and bi_area_ew_stats_2026_pm25.

Finally, we perform ensemble analysis that integrates all combinations of scenarios (A/B/C), models (RF/XGB), and pollutants (PM10/PM2.5). For each administrative dong and monitoring station, we compute how frequently it is classified as high-risk across these combinations. Based on this information, we create umd_pm_stats_2026_scenarios_with_classes.gpkg and umd_ensemble_2026_with_classes.gpkg, and use them in QGIS to produce maps of scenario-specific and ensemble risk levels for 2026 (Figures 3 and 4). Under PM10, all stations fall into a "low-risk" class, whereas under PM2.5, all but one station are classified as "very high risk."

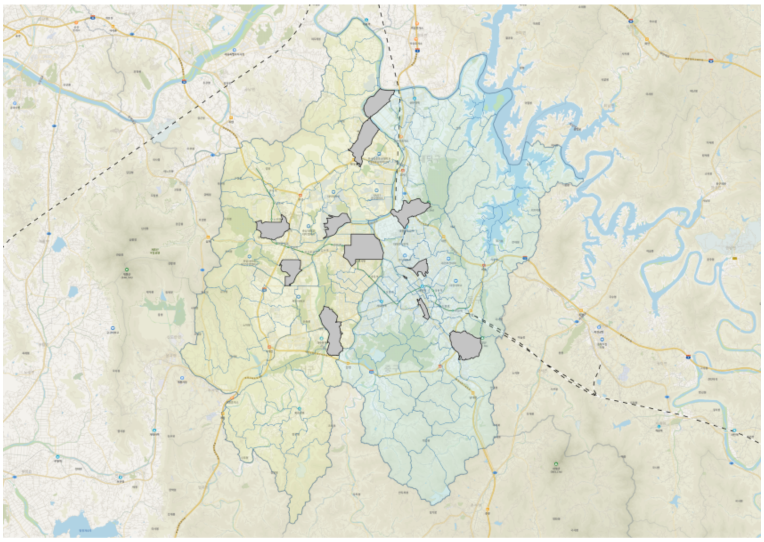


Figure 3. Ensemble risk map for PM10 in 2026 (PM10_2026_ensemble_risk).

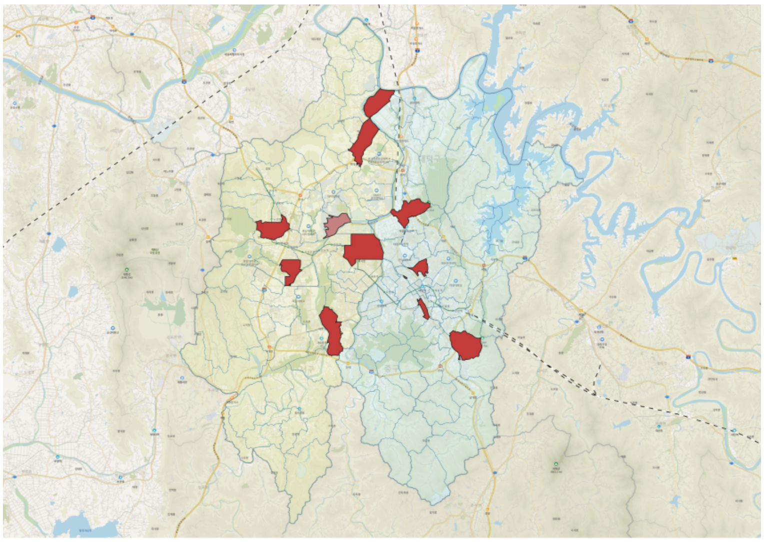


Figure 4. Ensemble risk map for PM2.5 in 2026 (PM2.5_2026_ensemble_risk).

4. Results

4.1 RQ1 – One-hour-ahead prediction performance

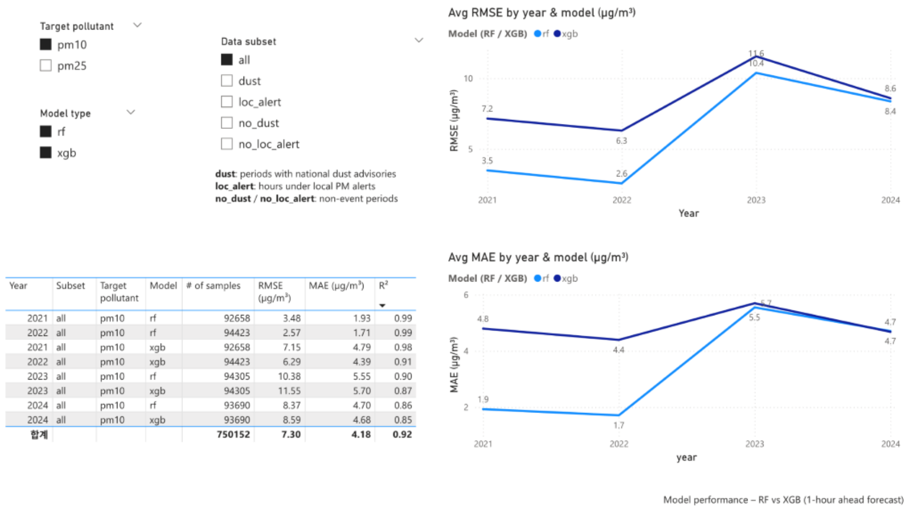
Prediction performance is evaluated using the "Model performance" pages (PM10, PM2.5) of the Power BI dashboard (Figures 5 and 6). Annual performance metrics for both models are computed based on the full 2021–2024 dataset.

For PM10, the Random Forest (RF) model achieves RMSE values of approximately 10.4 and 8.4 μg/m³ and MAE values of about 5.6 and 4.7 μg/m³ in the validation and external test years (2023 and 2024), respectively, with R^2 values of about 0.90 and 0.86. The XGBoost (XGB) model shows slightly larger errors over the same periods, with RMSE values of around 11.6 and 8.6 μg/m³ and R^2 values of 0.87 and 0.85.

For PM2.5, the RF model yields RMSE values of about 5.4 and 4.7 μg/m³ in 2023 and 2024, MAE values of 3.4 and 3.2 μg/m³, and R^2 values of 0.87 and 0.82, respectively. The XGB model exhibits similar levels of predictive performance, but overall the RF model maintains a slightly more stable pattern.

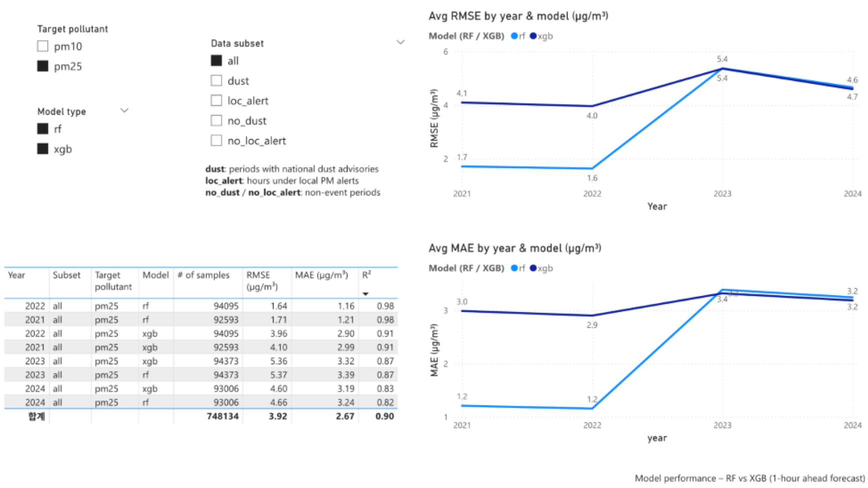
Both models show very high goodness-of-fit during the training period (2021–2022), with $R^2 \approx 0.98$–0.99, but RMSE and MAE increase for 2023–2024. In Figure 5, this is reflected in simultaneous upward trends in annual RMSE and MAE curves. This pattern suggests that year-to-year non-linearity in meteorological and emission conditions and the irregular nature of high-concentration events limit model generalization.

Furthermore, the same figures show increased error variance for samples associated with dust and loc_alert conditions, indicating that the models tend to partially underestimate peak concentrations during high-pollution or advisory periods. This implies that additional work is needed, such as assigning higher weights to event periods or enriching the feature set with local wind and humidity indicators, to improve performance for such episodes.



Year	Subset	Target pollutant	Model	# of samples	RMSE (μg/m³)	MAE (μg/m³)	R^2
2021	all	pm10	rf	92658	3.48	1.93	0.99
2022	all	pm10	rf	94423	2.57	1.71	0.99
2021	all	pm10	xgb	92658	7.15	4.79	0.98
2022	all	pm10	xgb	94423	6.29	4.39	0.91
2023	all	pm10	rf	94305	10.38	5.55	0.90
2023	all	pm10	xgb	94305	11.55	5.70	0.87
2024	all	pm10	rf	93690	8.37	4.70	0.86
2024	all	pm10	xgb	93690	8.59	4.68	0.85
합계				**750152**	**7.30**	**4.18**	**0.92**

Figure 5. Annual and condition-specific prediction performance for PM10.



Year	Subset	Target pollutant	Model	# of samples	RMSE (μg/m³)	MAE (μg/m³)	R^2
2022	all	pm25	rf	94095	1.64	1.16	0.98
2021	all	pm25	rf	92593	1.71	1.21	0.98
2022	all	pm25	xgb	94095	3.96	2.90	0.91
2021	all	pm25	xgb	92593	4.10	2.99	0.91
2023	all	pm25	xgb	94373	5.36	3.32	0.87
2023	all	pm25	rf	94373	5.37	3.39	0.87
2024	all	pm25	xgb	93006	4.60	3.19	0.83
2024	all	pm25	rf	93006	4.66	3.24	0.82
합계				**748134**	**3.92**	**2.67**	**0.90**

Figure 6. Annual and condition-specific prediction performance for PM2.5

4.2 RQ2 – Spatial vulnerability in 2021–2023

Spatial vulnerability over the three-year period is analyzed primarily using the "Overview" (Figure 7) and "Spatial patterns – by neighbourhood/season" (Figure 8) pages of the Power BI dashboard.

For the East/West comparison, the three-year mean PM10 concentration in the eastern sector (Dong-gu, Jung-gu, Daedeok-gu) is about 34–37 μg/m³, whereas the western sector (Seo-gu, Yuseong-gu) shows about 31–35 μg/m³, indicating generally higher levels in the east. For PM2.5, mean concentrations are approximately 17.3 μg/m³ in the eastern sector and 17.0 μg/m³ in the western sector, but the time fractions exceeding the standard are about 44% and 42%, respectively, again slightly higher in the east.

At the administrative-dong level (Figure 7), Gwanpyeong-dong, Munpyeong-dong, Noeun-dong, Dunsan-dong, Munchang-dong, and Jeongnim-dong maintain relatively high PM levels. These dongs largely overlap with areas that have high traffic volumes and a high share of commercial land use. Seasonal patterns are also evident: summer PM10 averages around 21 μg/m³, whereas winter values rise to 35–40 μg/m³, reflecting the influence of heating and stagnant atmospheric conditions.

The "Alerts & Dust" page (Figure 8) compares periods with local alerts and dust advisories. Even during alert episodes, the same administrative dongs tend to show high PM concentrations and longer exposure times, indicating that regions with high background levels remain more vulnerable during events. This pattern demonstrates that evaluating spatial risk based on "duration of exposure" rather than only on mean concentrations provides meaningful insights for urban management.

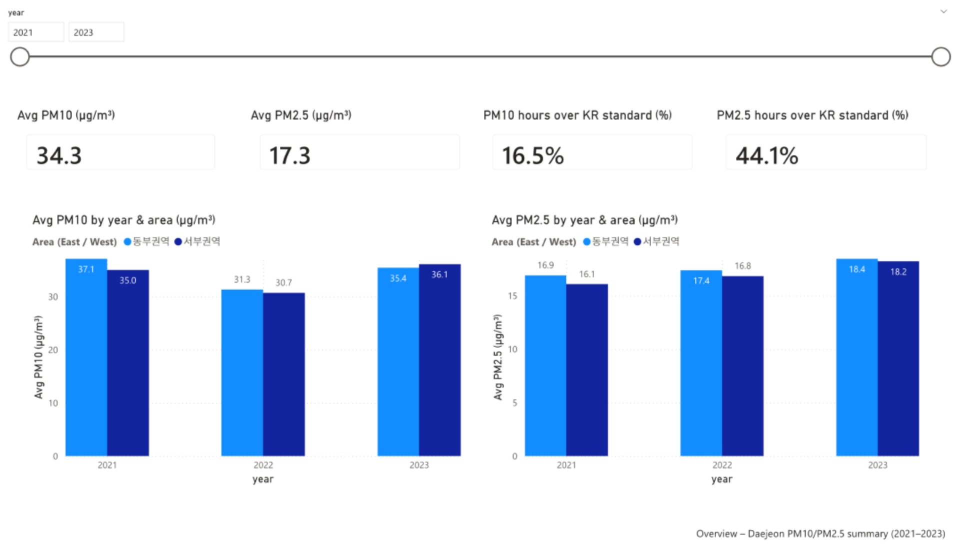


Figure 7. Overview of three-year mean PM levels and exceedance ratios by administrative dong and sector.

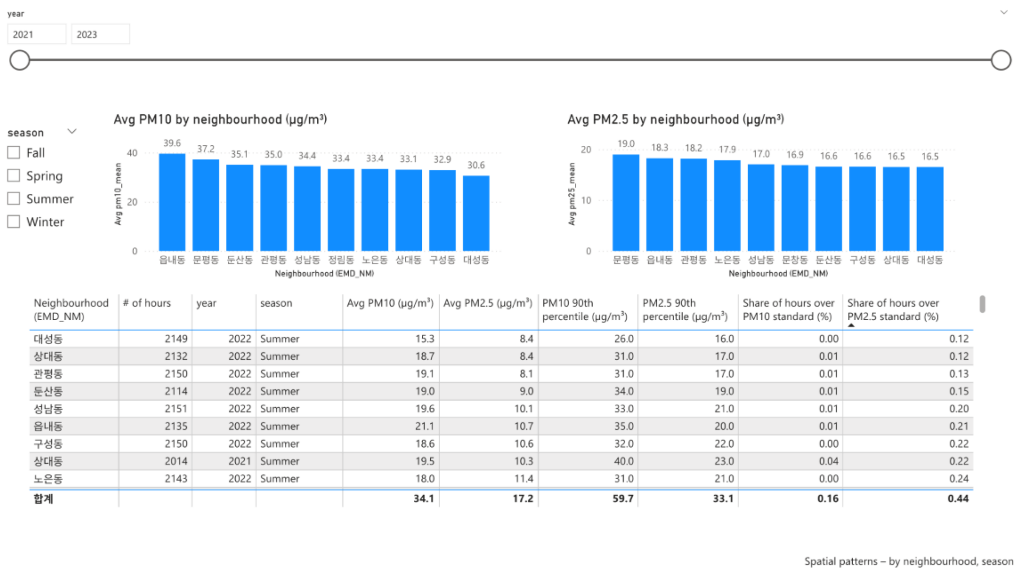


Figure 8. Spatial patterns of PM and exposure by neighbourhood and season.

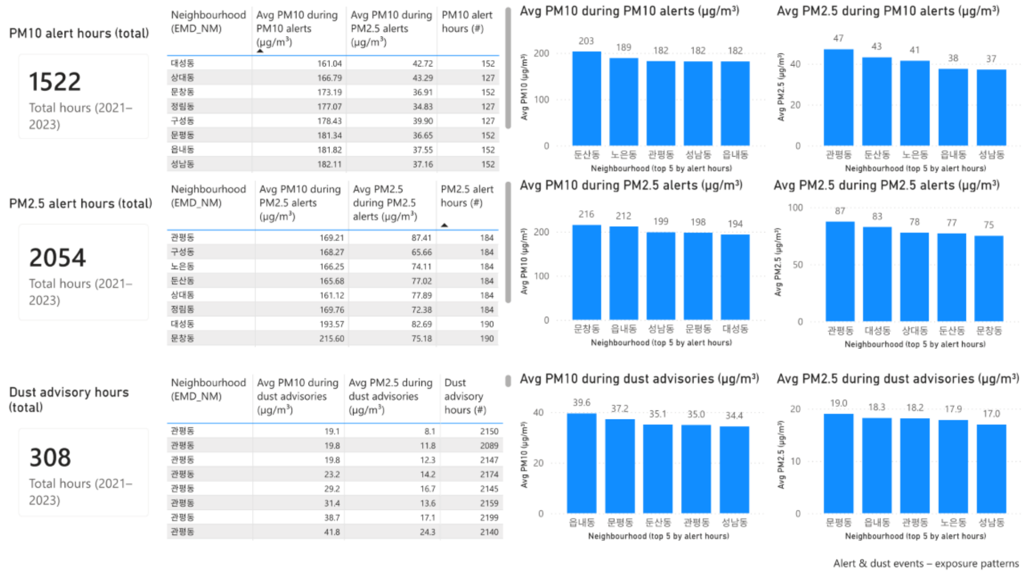


Figure 9. Exposure characteristics during local PM alerts and dust-advisory periods.

4.3 2026 multi-analog scenarios & ensemble risk

Near-future scenario analyses are based on the "Scenario 2026 – PM10/PM2.5" and "Ensemble (Part 2)" pages of the Power BI dashboard (Figures 10–12). Across scenarios A–C (2021–2023 analogs for 2026), RF and XGB results are similar, and spatial patterns are broadly consistent.

For PM10, the predicted 2026 mean concentrations are approximately 37 μg/m³ in the eastern sector and 35 μg/m³ in the western sector, indicating only minor differences. The fraction of hours exceeding 50 μg/m³ is generally around 10%, with relatively high exceedance frequencies in Munpyeong-dong, Eupnae-dong, and Seongnam-dong (Figure 10).

For PM2.5, mean concentrations are about 18 μg/m³ in both sectors, but dongs such as Munpyeong-dong, Gwanpyeong-dong, Guseong-dong, Dunsan-dong, and Jeongnim-dong show high exceedance ratios (≥25 μg/m³ for 20–30% of hours; Figure 11).

The ensemble results (Figure 12), which integrate three scenarios, two models, and two pollutants, indicate that all 11 urban air-monitoring stations fall into a high-risk group under the PM2.5 criterion, while no station is classified as high-risk under the PM10 criterion. In particular, Guseong-dong and Gwanpyeong-dong exhibit high-episode ratios of 66–100% across all scenarios, marking them as persistent hotspots in the Daejeon urban area.

These ensemble patterns are consistent with the current spatial distribution (2021–2023), where certain dongs in the eastern sector repeatedly show high exposure. Consequently, these areas should be prioritized in medium- and long-term air-quality management policies.

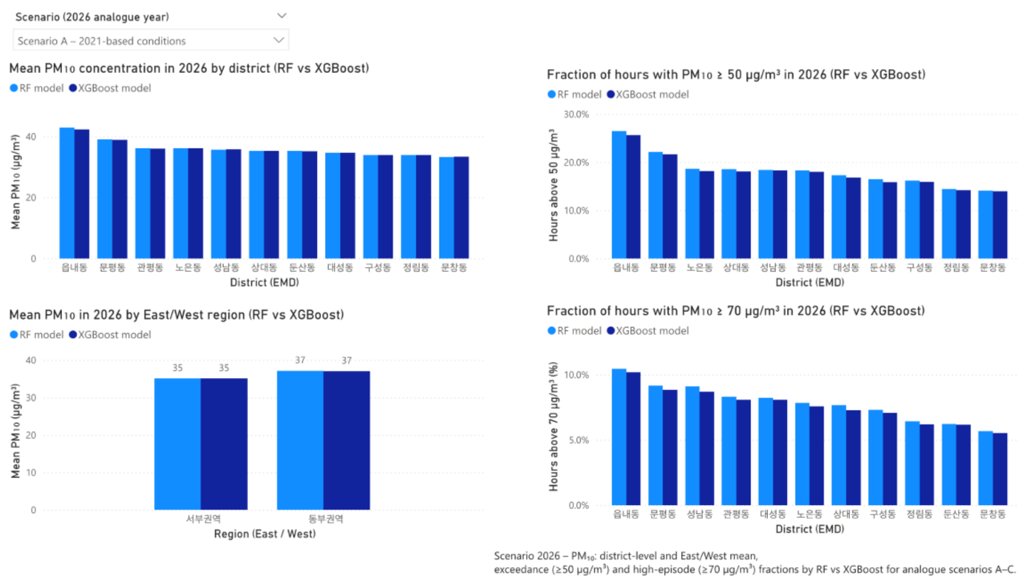


Figure 10. Scenario-based PM10 risk patterns in 2026.

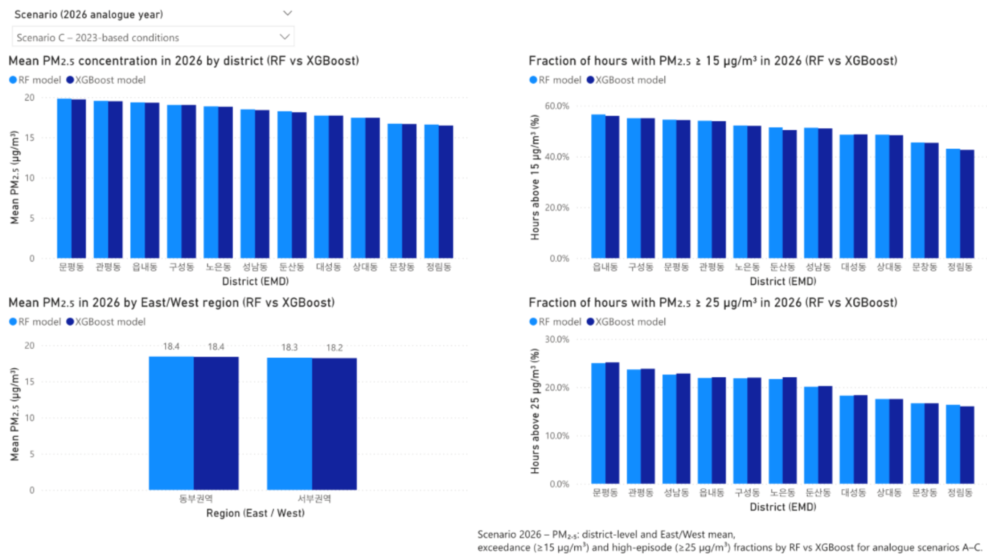


Figure 11. Scenario-based PM2.5 risk patterns in 2026.

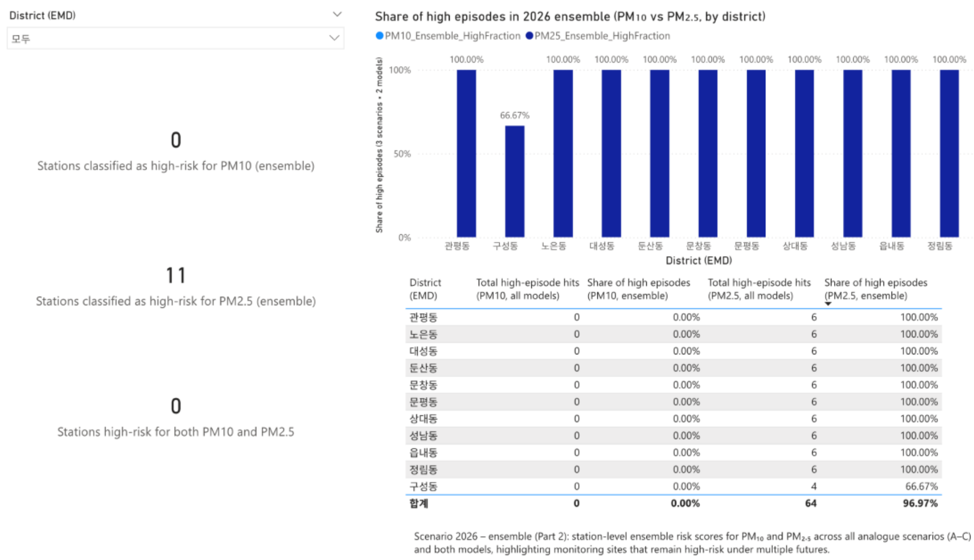


Figure 12. Ensemble risk patterns for 2026 across scenarios, models, and pollutants.

5. Discussion

5.1 Predictive capability and limitations

From the perspective of short-term prediction, the fact that publicly available monitoring data and simple time-series features can achieve R^2 values of roughly 0.8–0.9 for PM10 and PM2.5 in the validation and external test years (2023–2024) suggests that the proposed models can serve as lightweight, data-driven tools for city-level air-quality management. Despite being trained solely on data from 2021–2022, the models maintain reasonably high explanatory power for the subsequent two years, indicating that the feature set and model selection capture basic temporal and meteorological patterns effectively.

At the same time, the increased errors during local PM alerts, dust advisories, and high-concentration events highlight limitations of the current pipeline. High-pollution episodes are driven by a combination of factors, including free-tropospheric inflow, sudden increases in local emissions, and secondary formation processes. However, relatively simple time-series and basic meteorological features are insufficient to fully represent these non-linear mechanisms.

5.2 Spatial vulnerability and policy implications

The spatial vulnerability analysis indicates that both at present (2021–2023) and in the near future (2026), certain administrative dongs and sectors repeatedly exhibit high exposure. In particular, Gwanpyeong-dong, Munpyeong-dong, Noeun-dong, Dunsan-dong, Munchang-dong, and Jeongnim-dong show elevated mean concentrations and high fractions of high-concentration hours for both PM10 and PM2.5, and are repeatedly classified as high-risk in the 2026 multi-analog scenarios. These areas should be assigned high priority as long-term management targets.

Comparisons between East and West sectors reveal that the eastern sector generally has higher mean concentrations and exceedance ratios than the western sector. This likely reflects a higher density of old downtown, industrial, logistics, and traffic-intensive zones in the east. Future policy design should consider tailored mitigation strategies such as traffic demand management, promotion of low-emission vehicles, and localized emission reductions focusing on specific administrative dongs in the eastern sector.

The differences between PM10 and PM2.5 also have important policy implications. While no station is classified as high-risk under the PM10 ensemble criterion, all stations are high-risk under the PM2.5 criterion. This implies that, within the same city, a much broader area is structurally exposed to high risk from the perspective of fine particulate matter. Therefore, regulatory standards, alert system design, and health-impact assessments should give particular attention to PM2.5.

5.3 Strengths, limitations and future work

A major strength of this study is that it constructs a fully reproducible data–model–spatial-analysis pipeline using only public datasets from AirKorea, the Korea Meteorological Administration, and local alert and dust-advisory records. All processing steps and derived data structures are explicitly documented, enabling adaptation to other cities and periods with similar data availability. In addition, by integrating short-term prediction, spatial vulnerability analysis, and near-future scenarios in a single end-to-end framework, the study provides a systematic approach to identifying areas that are "currently high and likely to remain high" in terms of PM exposure.

The study also has several limitations. First, the analysis period is restricted to 2021–2024, which may be too short to fully capture the roles of climate variability and long-term emission trends. Second, the meteorological data rely on a single or small number of ASOS stations, potentially limiting the representation of local meteorological heterogeneity. Third, local alerts and dust advisories are defined using discrete categories determined by administrative agencies, which makes it difficult to fully capture continuous exposure intensity near threshold boundaries. Fourth, the absence of explicit features for emissions, traffic, and population exposure limits the ability to attribute high-concentration events to specific causes or to extend the analysis directly to health-impact assessment.

Future work should expand the analysis period and incorporate multiple meteorological stations or reanalysis datasets to better represent spatial variability in the meteorological field. Additional features related to traffic, industrial activity, and population distribution should be integrated to jointly analyze event causes and vulnerable populations. Hybrid modeling approaches that couple the current data-driven pipeline with physically based air-quality models (e.g., CMAQ, WRF-Chem) also hold promise. Finally, applying the same framework to other cities or regions and performing comparative analyses would provide broader insights into the spatial patterns of urban air-quality vulnerability in Korea.

Conclusion

This study presents a local case study for Daejeon Metropolitan City that integrates, within a single end-to-end pipeline based solely on public monitoring data, one-hour-ahead PM10/PM2.5 prediction, spatial vulnerability analysis for 2021–2023, and 2026 multi-analog scenarios with ensemble vulnerability assessment.

First, we confirm that using public PM and meteorological data and simple time-series features, one-hour-ahead PM10 and PM2.5 concentrations at 11 urban air-monitoring stations in Daejeon can be predicted with R^2 values of approximately 0.8–0.9 in the validation and external test years (2023–2024) (RQ1). This indicates that relatively simple data and feature combinations can provide practically useful levels of short-term predictive performance. At the same time, increased errors during local PM alerts, dust advisories, and high-concentration events highlight the need for additional features and model refinements to better capture high-pollution episodes.

Second, spatial vulnerability analysis for 2021–2023 at the administrative-dong and East/West sector levels shows that the eastern sector generally has higher mean concentrations and standard exceedance ratios than the western sector, and that several administrative dongs (Gwanpyeong-dong, Munpyeong-dong, Noeun-dong, Dunsan-dong, Munchang-dong, and Jeongnim-dong) exhibit structurally high-risk patterns for both PM10 and PM2.5 (RQ2). This suggests that differences in emissions, topography, meteorology, and urban structure lead to clearly differentiated spatial patterns of vulnerability within the city.

Third, multi-analog scenarios constructed by shifting the observed patterns in 2021–2023 to 2026, combined with RF/XGBoost models and ensemble analysis of PM10/PM2.5, reveal that some administrative dongs and sectors remain persistently high-risk across diverse near-future patterns (RQ3). In particular, all stations are classified as ensemble high-risk under the PM2.5 criterion, whereas no station is high-risk under the PM10 criterion, underscoring the need to prioritize PM2.5 in urban air-quality management.

In summary, this study proposes a reproducible analysis framework that integrates short-term prediction, spatial vulnerability, and near-future scenarios using only public monitoring data, and provides vulnerability assessments that jointly consider PM10 and PM2.5. The resulting framework offers a basis for future work in urban air-quality management and for more advanced modeling studies. Future research should focus on extending the temporal coverage, enriching the feature set, applying the framework to additional cities, and integrating physically based models and long-term climate/emission scenarios to further generalize and refine the approach.

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